

WEEK 4

Data Visualization Project

End-to-End Sales Analysis using Python, Excel and Power BI

Final Project Report

Prepared as part of the Week 4 internship assignment

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1. Project Overview

This project presents an end-to-end analysis of a business sales dataset. The workflow covered data inspection and cleaning in Python, analytical summaries in Pandas and Excel, and interactive visualization in Power BI. The final dashboard is organized into three analytical pages: Executive Overview, Product Intelligence, and Geography & Segment.

Objective

The objective was to transform a raw sales dataset into a clean analytical dataset, calculate useful business metrics, identify patterns across time, products, countries and segments, and communicate the results through an interactive dashboard.

Tools

Python, Pandas, NumPy, Matplotlib, Seaborn, Microsoft Excel and Microsoft Power BI.

2. Dataset & Data Preparation

The dataset contained 700 records and 16 original columns. The fields covered customer segment, country, product, discount band, units sold, pricing, gross sales, discounts, sales, COGS, profit and date-related fields.

Cleaning steps performed in the supplied Python notebook:

- Inspected shape, data types, descriptive statistics and missing values.
- Removed leading/trailing whitespace from column names.
- Checked for duplicate rows; no duplicates were identified.
- Filled missing numeric values with the median.
- Filled missing categorical values with the mode. The 53 missing Discount Band values were filled using the mode.
- Converted Date to datetime and derived Year, Month and Month_Name.
- Created Profit_Margin, Discount_Percentage and Revenue_Per_Unit calculated fields.

Data quality result

The cleaned analytical workbook contains 700 rows and 21 columns, including the added calculated fields. The cleaned CSV is included separately in the submission folder.

3. Analysis Methodology

The project follows a simple, traceable workflow from raw data to stakeholder-facing insights.

Stage	Activity	Purpose
01	Inspect	Understand columns, types, missing values and duplicates.
02	Clean	Standardize fields, handle missing values and validate the data.
03	Calculate	Create profit margin, discount percentage and revenue-per-unit metrics.
04	Analyze	Compare sales and profit by year, country, product and segment.
05	Visualize	Build supporting Excel charts and an interactive Power BI dashboard.
06	Communicate	Summarize findings and provide a reproducible submission package.

Calculated Metrics

- Profit Margin = Profit / Sales × 100
- Discount Percentage = Discounts / Gross Sales × 100
- Revenue per Unit = Sales / Units Sold
- YoY Growth = (Current Year Sales – Previous Year Sales) / Previous Year Sales

4. Business KPIs

The following KPIs were calculated from the cleaned dataset and used as the basis for the dashboard.

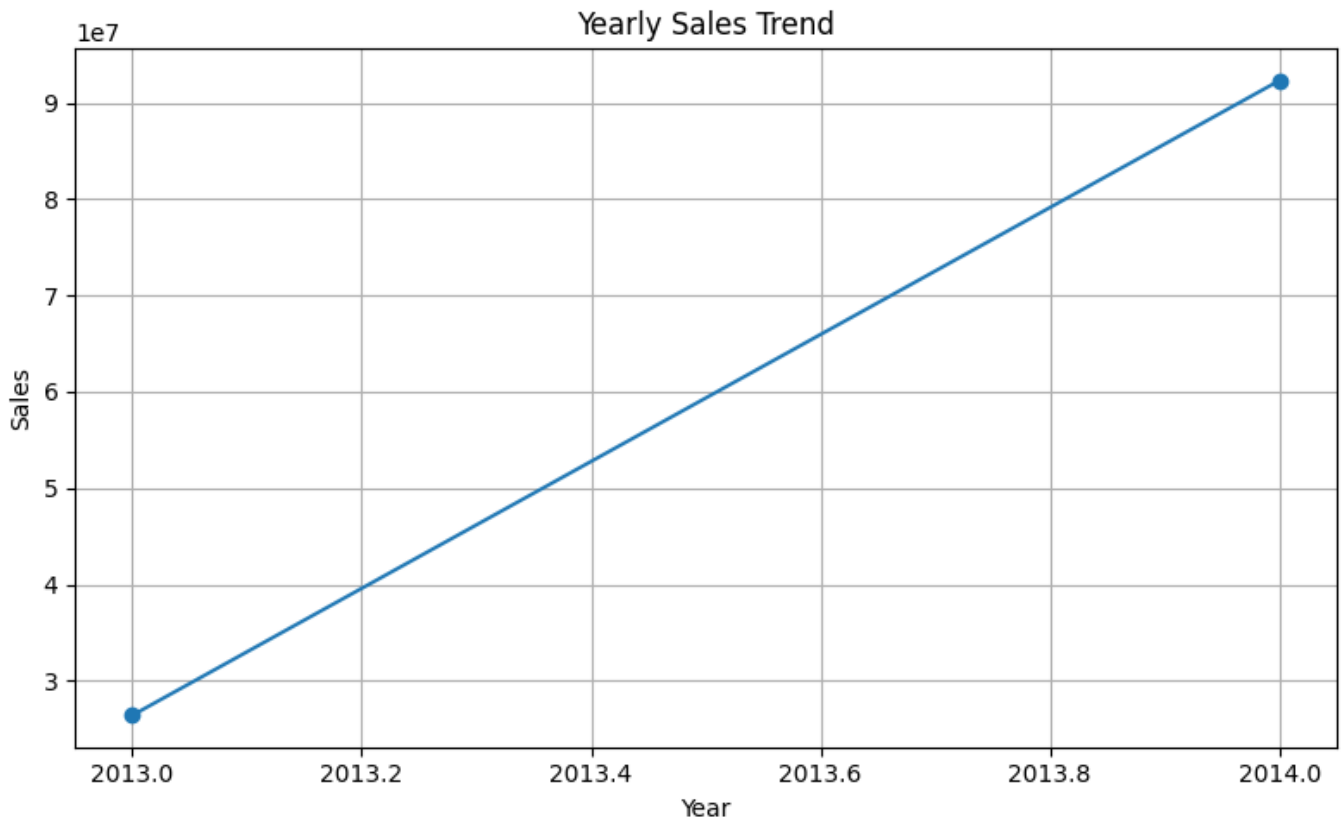
KPI	Result
Total Sales	\$118.73M
Total Profit	\$16.89M
Profit Margin	14.23%
Units Sold	1.13M
Average Revenue / Unit	\$105.46
YoY Growth %	249.46%
Discount Impact %	7.20%

These measures are designed to respond to dashboard filters, so the KPI values can be recalculated by year, country, segment or product in Power BI.

5. Key Findings & Business Insights

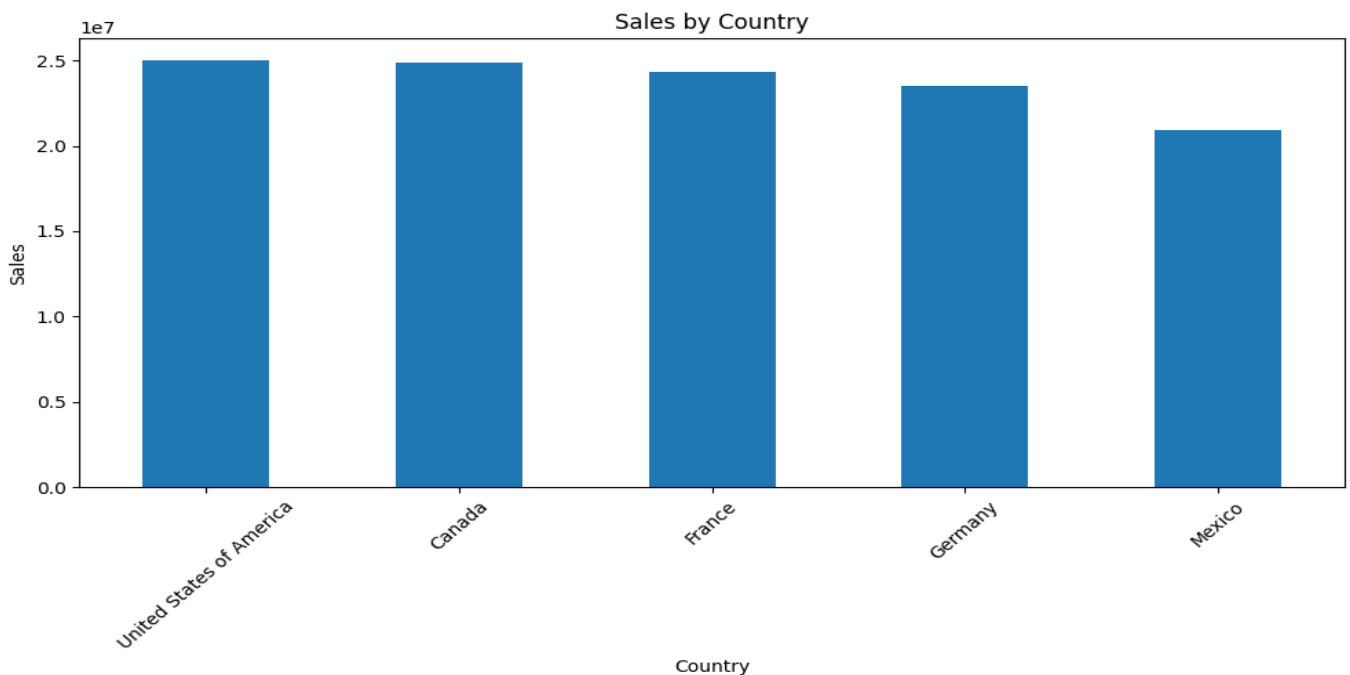
Yearly performance

Sales increased from approximately \$26.42M in 2013 to \$92.31M in 2014.



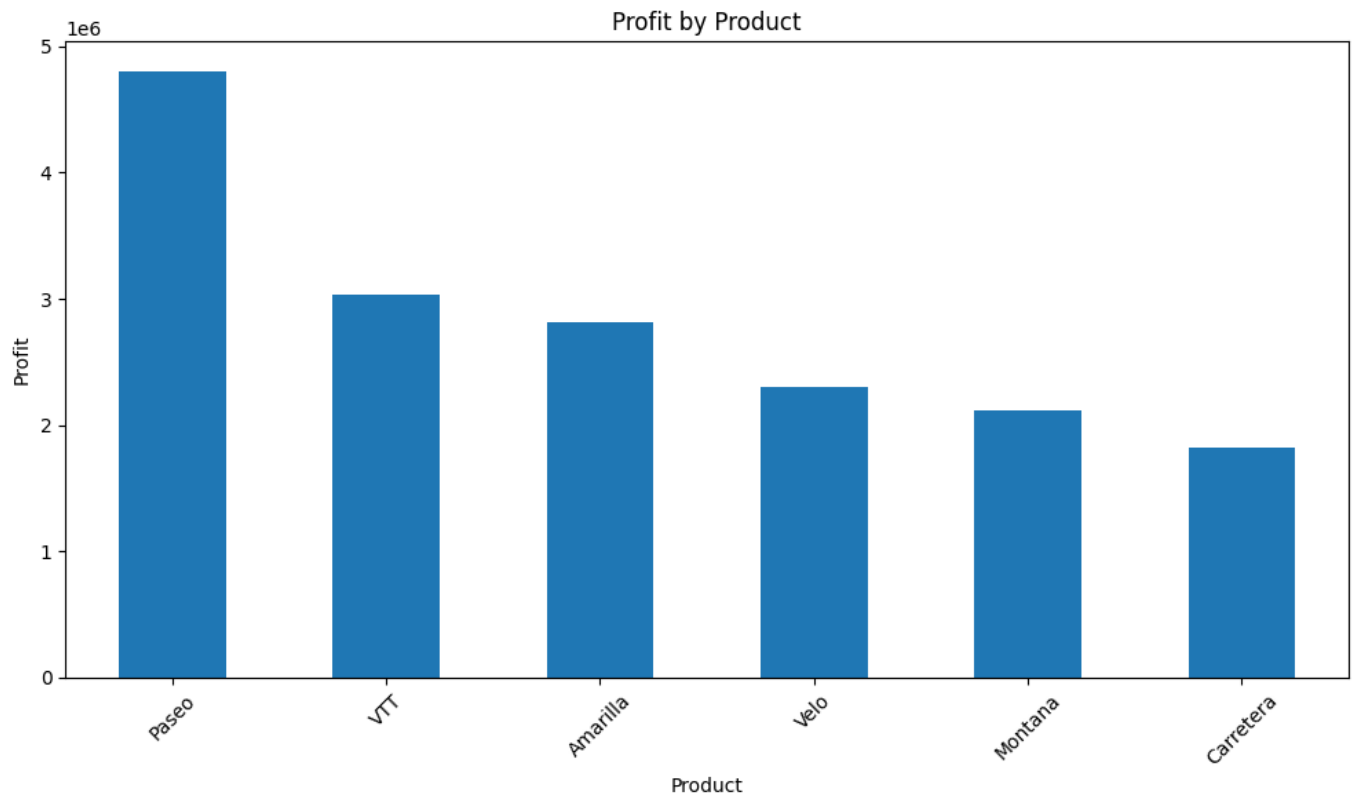
Country performance

The United States of America recorded the highest total sales, followed closely by Canada and France. The country comparison is visualized in the Power BI dashboard and supporting report chart.



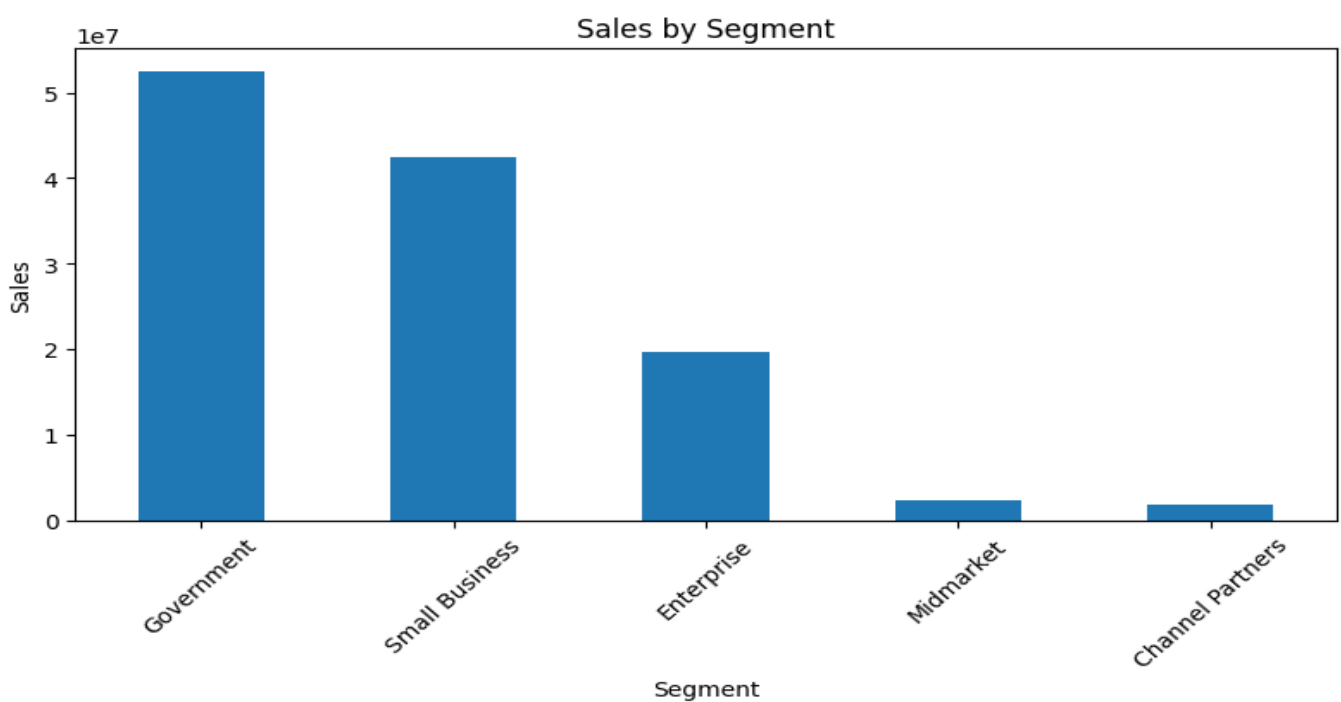
Product performance

Paseo generated the highest total sales and the highest total profit among the six products in the dataset. VTT and Amarilla were also significant contributors to total profit.



Segment performance

Government generated the highest total sales and profit. Small Business was the second-largest sales segment. Enterprise recorded a negative total profit in the analyzed dataset, which is an important point for further investigation.



6. Power BI Dashboard

The Power BI dashboard is included as a .pbix file in the submission package. The dashboard is organized into three pages to separate executive-level KPIs from product and geographic analysis.

Page 1 – Executive Overview

Designed to provide a quick business snapshot using KPI cards, sales and profit trend analysis, country performance, product contribution, segment mix and interactive slicers.

Page 2 – Product Intelligence

Focuses on product-level sales and profit performance, product contribution, distribution analysis and the relationship between sales and profit.

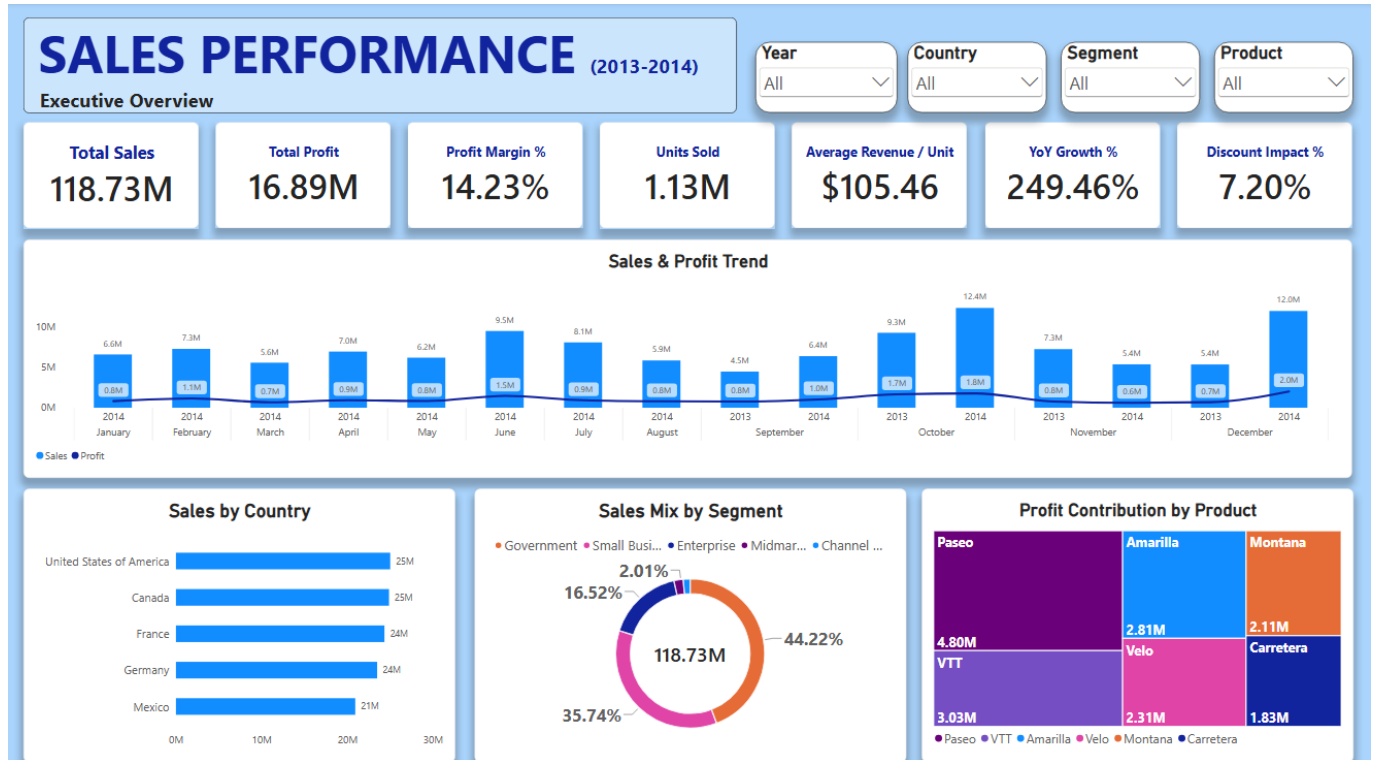
Page 3 – Geography & Segment

Focuses on country-level sales and profit, segment performance and discount-related analysis.

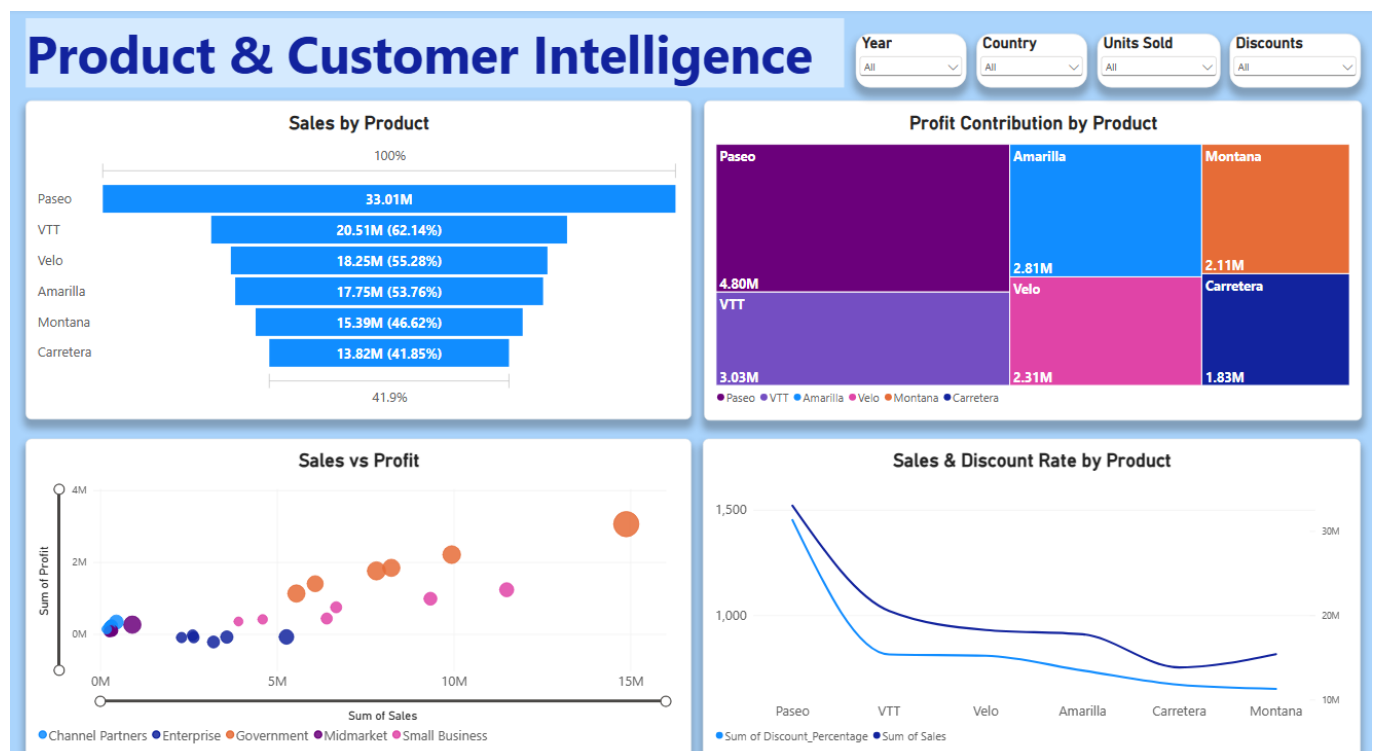
The dashboard uses slicers so stakeholders can explore the data by year, country, segment and product.

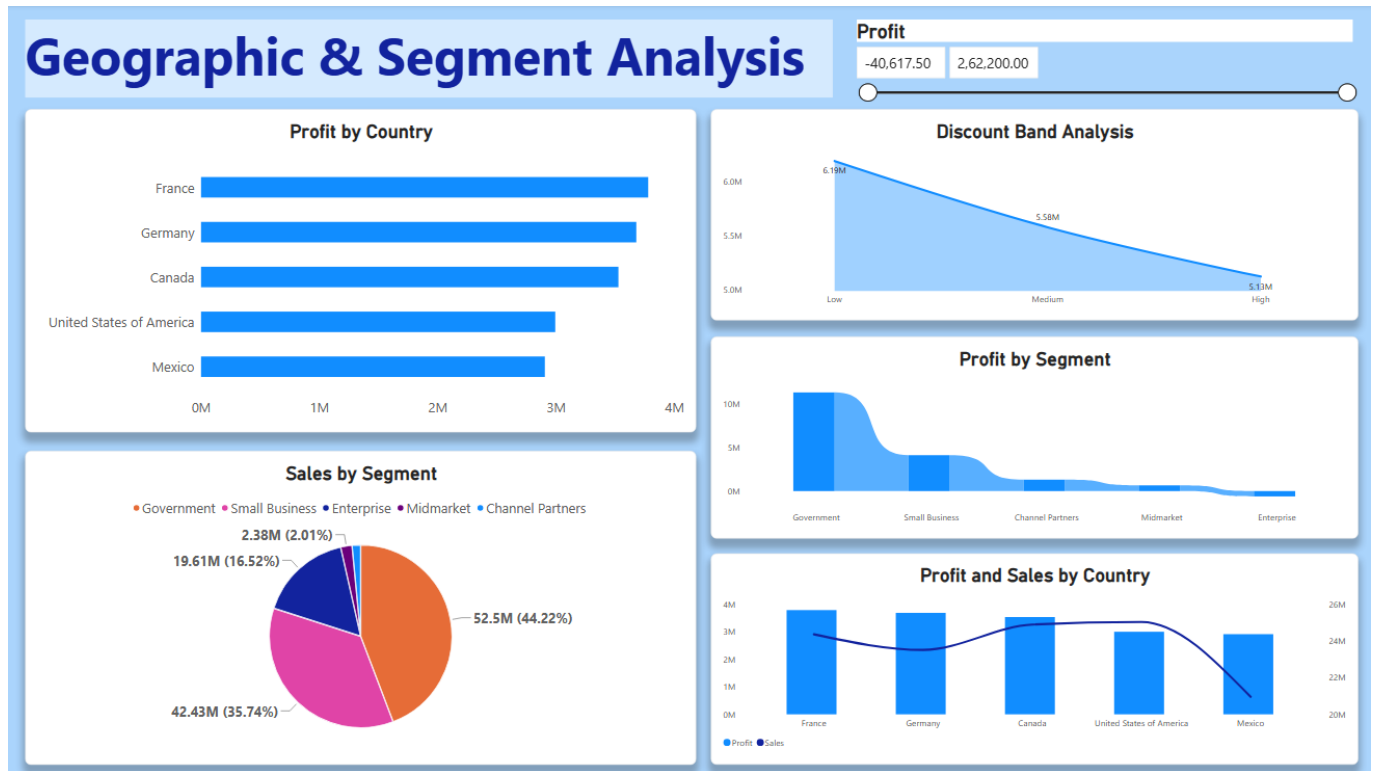
7. Dashboard Screenshots

Page 1 – Executive Overview



Page 2 – Product & Customer Intelligence





8. Code Appendix

1. IMPORT LIBRARIES

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2. LOAD DATASET

```
df = pd.read_csv("Sample data.csv")
df.head()
```

3. CHECK DATASET SHAPE

```
df.shape
```

4. CHECK DATA INFORMATION

```
df.info()
```

5. DESCRIPTIVE STATISTICS

```
df.describe()
```

6. CHECK COLUMN NAMES

```
df.columns
```

7. CHECK MISSING VALUES

```
df.isnull().sum()
```

8. CLEAN COLUMN NAMES

```
df.columns = df.columns.str.strip()
df.columns
```

9. CHECK DUPLICATE ROWS

```
df.duplicated().sum()
```

10. REMOVE DUPLICATE ROWS

```
df = df.drop_duplicates()
```

11. VERIFY DUPLICATE ROWS

```
df.duplicated().sum()
```

12. CHECK MISSING VALUES AGAIN

```
df.isnull().sum()
```

13. HANDLE MISSING NUMERIC VALUES

```
numeric_columns = df.select_dtypes(include=np.number).columns
for col in numeric_columns:
    df[col] = df[col].fillna(df[col].median())
```

14. HANDLE MISSING CATEGORICAL VALUES

```
categorical_columns = df.select_dtypes(include="object").columns
for col in categorical_columns:
    df[col] = df[col].fillna(df[col].mode()[0])
```

15. VERIFY MISSING VALUES

```
df.isnull().sum()
```

16. CONVERT DATE COLUMN

```
df["Date"] = pd.to_datetime(  
    df["Date"],  
    errors="coerce"  
)
```

17. CHECK DATE VALUES

```
df["Date"].head()
```

18. CREATE DATE-BASED COLUMNS

```
df["Year"] = df["Date"].dt.year  
df["Month"] = df["Date"].dt.month  
df["Month_Name"] = df["Date"].dt.month_name()
```

19. CREATE PROFIT MARGIN

```
df["Profit_Margin"] = (  
    df["Profit"] / df["Sales"]  
) * 100
```

20. CREATE DISCOUNT PERCENTAGE

```
df["Discount_Percentage"] = (  
    df["Discounts"] / df["Gross Sales"]  
) * 100
```

21. CREATE REVENUE PER UNIT

```
df["Revenue_Per_Unit"] = (  
    df["Sales"] / df["Units Sold"]  
)
```

22. VIEW CLEANED DATA

```
df.head()
```

23. CALCULATE OVERALL BUSINESS METRICS

```
total_sales = df["Sales"].sum()  
total_profit = df["Profit"].sum()  
total_units = df["Units Sold"].sum()  
total_gross_sales = df["Gross Sales"].sum()  
total_discounts = df["Discounts"].sum()
```

24. CALCULATE OVERALL PROFIT MARGIN

```
overall_profit_margin = (  
    total_profit / total_sales  
) * 100  
print(  
    "Overall Profit Margin:",  
    overall_profit_margin  
)
```

25. SALES BY COUNTRY

```
sales_by_country = (  
    df.groupby("Country")["Sales"]  
    .sum()  
    .sort_values(ascending=False)  
)  
sales_by_country
```

26. PROFIT BY COUNTRY

```
profit_by_country = (  
    df.groupby("Country")["Profit"]  
    .sum()  
    .sort_values(ascending=False)  
)  
profit_by_country
```

27. SALES BY PRODUCT

```
sales_by_product = (  
    df.groupby("Product")["Sales"]  
    .sum()  
    .sort_values(ascending=False)  
)  
sales_by_product
```

28. PROFIT BY PRODUCT

```
profit_by_product = (  
    df.groupby("Product")["Profit"]  
    .sum()  
    .sort_values(ascending=False)  
)  
profit_by_product
```

29. SALES BY SEGMENT

```
sales_by_segment = (  
    df.groupby("Segment")["Sales"]  
    .sum()  
    .sort_values(ascending=False)  
)  
sales_by_segment
```

30. PROFIT BY SEGMENT

```
profit_by_segment = (  
    df.groupby("Segment")["Profit"]  
    .sum()  
    .sort_values(ascending=False)  
)  
profit_by_segment
```

31. YEARLY SALES

```
yearly_sales = df.groupby("Year")["Sales"].sum()  
yearly_sales
```

32. YEARLY PROFIT

```
yearly_profit = df.groupby("Year")["Profit"].sum()  
yearly_profit
```

33. MONTHLY SALES

```
monthly_sales = (  
    df.groupby(["Year", "Month"])["Sales"]  
    .sum()  
    .reset_index()  
)  
monthly_sales.head()
```

34. VISUALIZATION – SALES BY COUNTRY

```
plt.figure(figsize=(10,6))
sales_by_country.plot(kind="bar")

plt.title("Sales by Country")
plt.xlabel("Country")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.tight_layout()\
plt.show()
```

35. VISUALIZATION – PROFIT BY PRODUCT

```
plt.figure(figsize=(10,6))
profit_by_product.plot(kind="bar")
plt.title("Profit by Product")
plt.xlabel("Product")
plt.ylabel("Profit")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

36. VISUALIZATION – SALES BY SEGMENT

```
plt.figure(figsize=(8,5))
sales_by_segment.plot(kind="bar")
plt.title("Sales by Segment")
plt.xlabel("Segment")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

37. VISUALIZATION – YEARLY SALES TREND

```
plt.figure(figsize=(8,5))
yearly_sales.plot(
    kind="line",
    marker="o"
)
plt.title("Yearly Sales Trend")
plt.xlabel("Year")
plt.ylabel("Sales")
plt.tight_layout()
plt.show()
```

38. SAVE CLEANED DATASETdf.to_csv(

```
    "cleaned_sales_data.csv",
    index=False
)
```

9. Conclusion

The project demonstrates a complete data-analysis workflow from raw data preparation to business-facing visualization. Python was used to inspect and clean the data and create calculated metrics; Excel provided supporting analysis; and Power BI was used to communicate the results interactively.

The analysis shows total sales of approximately \$118.73M and total profit of approximately \$16.89M, with an overall profit margin of approximately 14.23%. Sales grew substantially between 2013 and 2014, while product, country and segment analysis revealed meaningful differences in contribution and profitability.

The final project files are organized so that the reviewer can trace the workflow from the original dataset through Python, cleaned data, Excel analysis, Power BI dashboard and final report.

10. Submission File Structure

Week 4 Data Visualization Power BI/

- ├── README.md
- ├── 01 Dataset/
 - └── Sample data.csv
- ├── 02 Python/
 - └── Week_4_Data_Analysis.ipynb
- ├── 03 Cleaned_Data/
 - └── Cleaned Sales Data.csv
- ├── 04 Excel/
 - └── Week_4_Analysis.xlsx
- ├── 05 PowerBI/
 - └── Week 4 Data Visualization Power BI.pbix
- ├── 06 Screenshots/
 - └── Dashboard 1.png
 - └── Dashboard 2.png
 - └── Dashboard 3.png
- └── 07 Report/
 - └── Week_4_Data_Visualization_Report.pdf